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Artificial Intelligence

Automated Segmentation and Characterization of Retinal Hyperreflective Foci in Age-Related Macular Degeneration

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Purpose: Retinal hyperreflective foci (HRF) are associated with higher risk of progression to advanced age-related macular degeneration (AMD). To enable automated detection, we developed Foci-Net, a convolutional neural network for segmenting HRF in optical coherence tomography (OCT) volumes.

Methods: This cross-sectional study included eyes clinically diagnosed with intermediate to advanced AMD and healthy controls. We acquired 6 × 6-mm macular-centered scans using 2 OCT systems. Foci-Net modifies the U-Net by replacing the bottleneck with a fine-to-coarse feature extraction block to improve segmentation of both small and large foci. HRF volume was quantified as the proportion of total volume within each retinal layer and characterized by anatomic location. Model performance was evaluated at the count, pixel, eye, and B-scan levels using F1-score, area under the curve (AUC) of the receiver operating characteristic (ROC) curve, precision, sensitivity, and specificity.

Results: Sixty-one volumetric OCT scans (50 AMD and 11 healthy control eyes) obtained from 50 participants, age 77.65 ± 9.24 years (mean ± SD), of whom 72% were women. Foci-Net achieved F1-scores of 73.0 ± 3.8% counts (based on en face analysis) and 71.6 ± 29.1% pixels (based on cross-sectional analysis) level relative to the ground truth in 3-fold cross-validation. Diagnostic performance was strong, with an AUC of 100 ± 0.0% (eye) and 92.7 ± 0.3% (B-scan) level. HRF was least (2.10%) abundant in the ganglion cell complex and most (55.37%) in the outer nuclear layer.

Conclusions: A deep learning approach enables fast HRF segmentation, supporting its role as a biomarker for predicting AMD progression.

Translational Relevance: The proposed HRF segmentation method could enhance AMD diagnosis and prognosis.

Introduction

Age-related macular degeneration (AMD) is the primary cause of irreversible blindness and vision loss.[1,2] Hyperreflective foci (HRF) are a key feature visible with optical coherence tomography (OCT) that can help in understanding the risk of progression from early to late-stage AMD.[3–6] HRF are highly reflective, well-circumscribed, dot-like lesions that appear on OCT with similar or higher reflectivity to the retinal pigment epithelium (RPE).[7–9] These foci are believed to represent microglia cells and migrated RPE cells that have separated from the RPE layer.[3,10] The quantity of HRF in participants with intermediate AMD (iAMD) significantly increases the risk of progression to late-stage AMD and the development of geographic atrophy.[7] Therefore, accurate automated identification and quantification of these foci may enhance risk stratification for vision loss in patients with AMD.

Several studies have used manual annotation and automated approaches to segment HRF in AMD. Echols et al.[8] manually counted HRF and hyperreflective specks (HRS) to examine the correlation with rod- and cone-mediated vision. They found that more

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HRF and HRS were strongly associated with delayed rod-mediated dark adaptation ($P < 0.0001$) and with worse low-luminance visual acuity ($P = 0.0117$). Duic et al.[3] used manual grading of HRF (binary presence/absence) to understand their distribution across different AMD severities. They observed that HRF presence increased with AMD severity and was associated with poorer rod-mediated visual function and greater disruption of retinal structure. However, manual segmentation is variable between graders, time-consuming, and resource-intensive, making it challenging to obtain large datasets. Bogunović et al.[11] developed an automated voxel-based method for HRF segmentation using principal component analysis (PCA) for feature extraction and a random forest classifier to estimate pixel-level probabilities. Methods that rely on handcrafted features often miss the local and global context, reducing their ability to generalize effectively. Schlegl et al.[12] used an automated deep learning approach using a ResUnet architecture, applied to a dataset consisting of cases with AMD, retinal vein occlusion (RVO), and diabetic macular edema (DME). Goel et al.[9] utilized a U-Net with pre-trained ResNet34 to segment HRF in a dataset comprising subjects with AMD and diabetic retinopathy (DR). These methods greatly improve HRF segmentation accuracy using deep learning approaches. However, limitations remain regarding false positive results when using small and hybrid datasets. HRF may have varying causes and morphologies across different retinal diseases. They are linked to hard exudates in DR and may appear in different retinal locations.[13] Training a model on reliable gold-standard labels is the key step required in clinical applications to prevent bias toward disease-specific patterns.

In this study, we propose an enhanced deep convolutional neural network (CNN), named Foci-Net, for HRF segmentation in AMD. The key contributions are (1) introducing a fine-to-coarse feature extraction (FCFE) block in the network's bottleneck to capture features at various scales, (2) characterizing HRF across retinal layers, and (3) providing 2-dimensional (2D) and 3-dimensional (3D) visualizations of HRF.

Methods

Data Acquisition

This study was approved by the Oregon Health & Science University institutional review board, with informed consent obtained from all participants, adhering to the Declaration of Helsinki. Healthy eyes and eyes diagnosed with intermediate to advanced stages of AMD through clinical examination were included in this study. Eye selection was based on confirmed AMD diagnosis and scan quality. In unilateral cases, the affected eye was included. In bilateral cases, both eyes were included when scan quality was adequate; otherwise, the eye with inadequate scan quality was excluded. The 6 × 6-mm scans centered on the macula were obtained using 2 commercial spectral domain OCT systems: (1) Solix, 120-kHz system (Visionix/Optovue Inc., Fremont, CA) with a central wavelength of 840-nm, from which we obtained 44 volumetric OCT scans. (2) RTVue-XR, 70-kHz system (Visionix/Optovue Inc., Fremont, CA) with 840-nm central wavelength, from which we obtained 17 volumetric OCT scans. The Solix system generates 6.4 × 6.4-mm OCT volumes sampled at 512 A-scans per B-scan (X) and 512 B-scans per volume (Y), with each B-scan having pixel dimensions of 640 (Z) × 512 (X). In contrast, the RTVue-XR system generates 6 × 6-mm volumes sampled at 400 A-scans per B-scan (X) and 400 B-scans per volume (Y), with each B-scan having pixel dimensions of 640 (Z) × 400 (X). The CNN model takes individual B-scans as input and outputs a corresponding segmentation mask for each B-scan. Therefore, consistent per–B-scan input dimensions were required. To ensure input standardization across devices, Solix and RTVue-XR B-scans were cropped to the retinal region to exclude excess vitreous background, yielding a uniform B-scan size of 400 (Z) × 400 (X) pixels, whereas the number of B-scans per volume (slow-axis sampling) remained unchanged.

Data Preprocessing

To enhance image quality and reduce speckle noise, we applied image preprocessing techniques, including contrast stretching and non-local means filtering in MATLAB, while ensuring the preservation of fine structural details, as shown in Figure 1. Two trained graders manually delineated HRF areas to generate a ground truth mask. Graders identified HRF as a well-circumscribed, hyperreflective lesion that is clearly separate from background retinal structures. The gradings were performed on individual B-scans in each volumetric OCT scan, creating a gold standard HRF mask (see Fig. 1B, green). Both graders examined questionable B-scans, and a common agreement was achieved by discussion. The retinal layers were segmented using a guided bidirectional graph search algorithm.[14] Retinal layer corrections and HRF delineation were manually performed using our COOL-ART software, developed by Dr. Jia and her team at the Center for Ophthalmic Optics and Lasers laboratory.

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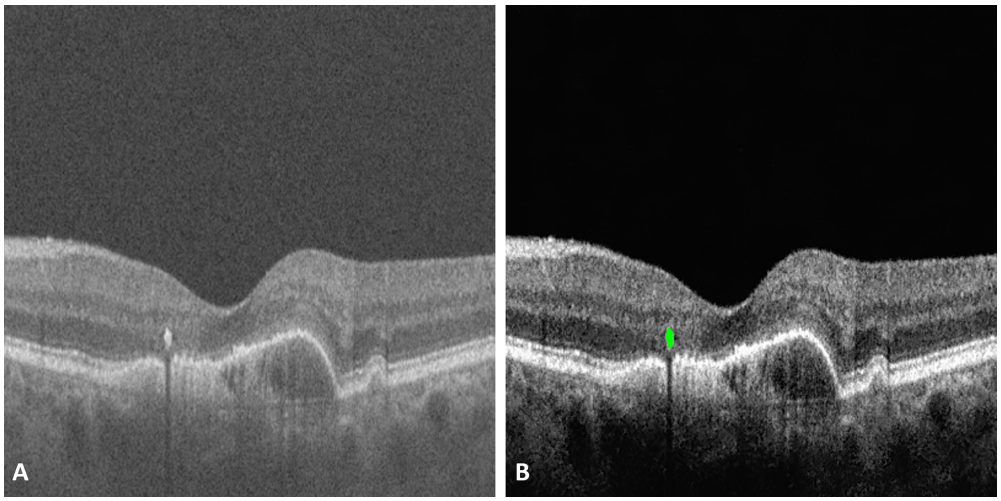


Figure 1. Representative example of image preprocessing with hyperreflective foci (HRF) delineation. (**A**) Original OCT B-scan. (**B**) Graded HRF annotations (*green*) overlaid on the processed OCT B-scan.

Designing a Convolutional Neural Network

Deep learning methods, such as CNNs, can automatically learn features, surpassing the limitations of handcrafted feature-based methods.[15] Complex tasks like HRF segmentation require global contextual and local spatial information preservation. U-Net type CNNs perform both downsampling to capture global contextual information and upsampling to reconstruct the image while incorporating skip connections to preserve local spatial details.[16] We adopted a U-Net type CNN and designed Foci-Net (Fig. 2A) with modifications to segment HRF. For better generalization, the modification includes replacing normal convolution with residual convolution blocks (Fig. 2B, ResBlock) to address the vanishing gradient problem.[17] An FCFE block (Fig. 2C), motivated by the GoogLeNet Inception module,[18] was placed in the bottleneck of the network to extract features at various scales. This block increases the ability of the model to segment foci of varying sizes.[19–21]

The source code for this model is available at https://github.com/octangio/FociNet.

Loss Function

To achieve both pixel and boundary-level precision in HRF segmentation, we used a combined binary cross-entropy (BCE) and Dice loss function[22]:

$$\mathrm{Loss_{Bce}} = -\frac{1}{N}\sum_{n=1}^{N}\left(y_n \log\left(p_n\right) + \left(1 - y_n\right)\log\left(1 - p_n\right)\right) \quad (1)$$

$$\mathrm{Loss_{Dice}} = \frac{2\left(\sum_{n=1}^{N} p_n y_n\right) + s}{\sum_{n=1}^{N} p_n + \sum_{n=1}^{N} y_n + s} \quad (2)$$

$$\mathrm{Loss_{combined}} = a\mathrm{Loss_{Bce}} + b\mathrm{Loss_{Dice}} \quad (3)$$

where y_n represents the ground truth (gold standard) for pixel n and p_n is the Foci-Net prediction for the same pixel. N is the total number of pixels. Additionally, s is a smoothing constant to avoid zero division. During training setting different values of a and b, it was found that assigning a higher weight to the Dice loss resulted in improved performance. The final combined loss function ($Loss_{combined}$), is a weighted combination of BCE and Dice loss, with weighting factors $a = 0.5$ for BCE loss ($Loss_{Bce}$) and $b = 1$ for the Dice loss ($Loss_{Dice}$).

Training Process

To prevent data leakage and ensure that scans from the same patient do not appear in both training and testing sets, all volumetric OCT scans were randomly split at the patient level into training, validation, and testing subsets. Both training and validation sets contained equal numbers of HRF and no-HRF B-scans. A three-fold cross-validation with non-overlapping test and validation sets was used to ensure diverse evaluation. The batch size was set to 8, with an initial learning rate of 0.0005. Early stopping (patience of 20 epochs) was applied within a total of 50 epochs to prevent overfitting. The Adam optimizer minimized the combined loss function,[23] while a reduce on plateau scheduler dynamically adjusted the learning rate when the validation loss plateaued, improv-

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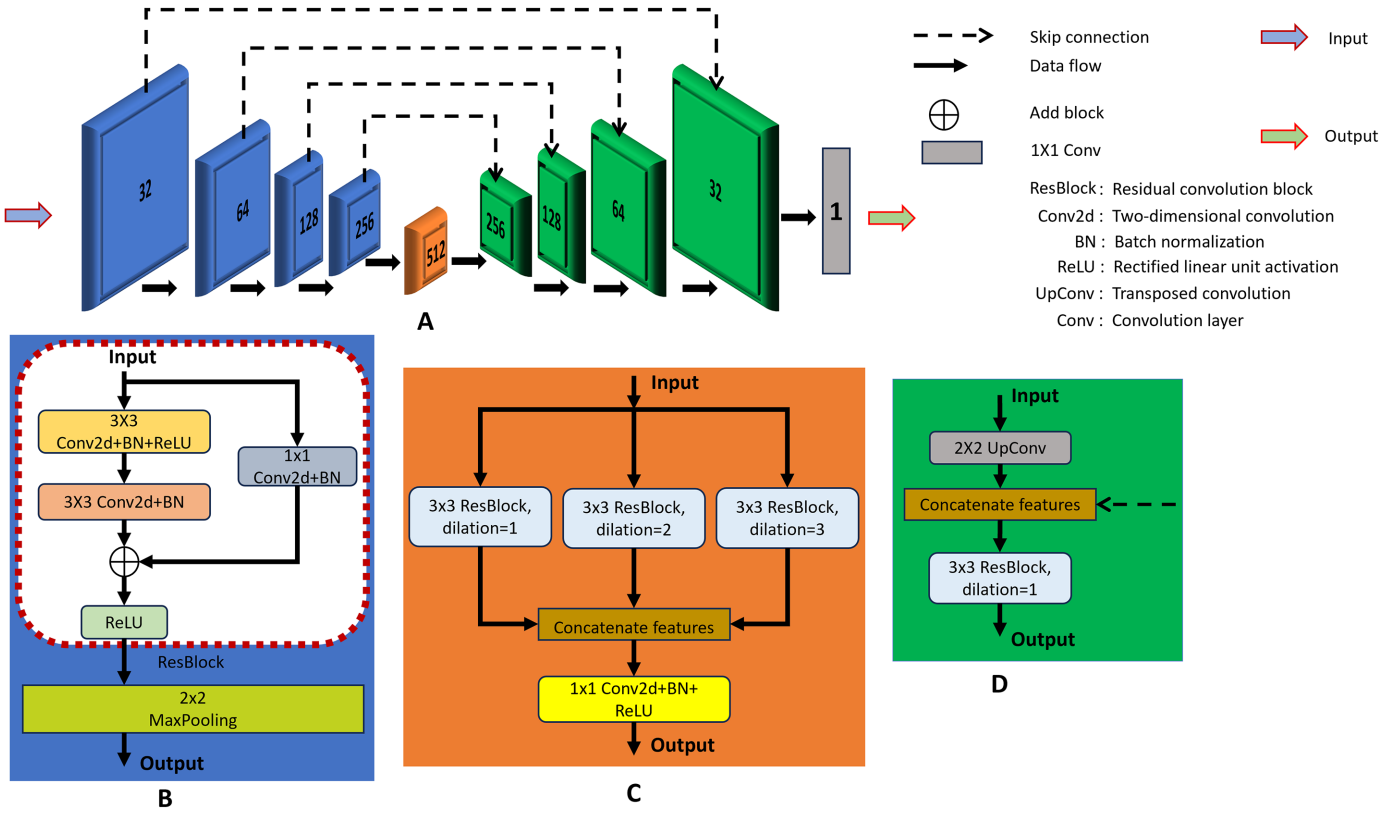


Figure 2. Architecture of the proposed deep convolutional neural network. (**A**) Foci-Net architecture with the number of output feature channels at each layer indicated (32, 64, 128, 256, and 512). (**B**) Encoder block. (**C**) Fine-to-coarse feature extraction (FCFE) block in the bottleneck to enhance multi-scale feature extraction and improve segmentation of small and large hyperreflective foci. (**D**) Decoder block.

ing model generalization. Data augmentation included horizontal flipping, and TensorBoard was utilized to monitor and tune model parameters.

Foci-Net was implemented in Python 3.10 using the PyTorch 2.2.1 framework on an Ubuntu 20.04 server, equipped with an Intel Xeon Gold 6254 CPU (3.10 GHz), two GeForce RTX 3090 GPUs, and 512 GB of RAM.

Performance Evaluation

To evaluate the performance of our proposed model on both segmentation and classification tasks, we calculated the F1-score, precision, sensitivity, specificity, and area under the curve (AUC) of the receiver operating characteristic (ROC) curve, as defined below:

$$\mathrm{F1} = \frac{2 \times \mathrm{TP}}{2 \times \mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \quad \mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}},$$
$$\mathrm{Sensitivity} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \quad \mathrm{Specificity} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \quad (4)$$

Where true positives (TP) are the number of correctly identified positives, true negatives (TN) are the number of correctly identified negatives, false positives (FP) represent incorrectly identified positives, and false negatives (FN) represent incorrectly identified negatives. Paired statistical comparisons between models were performed using DeLong's test for correlated ROC curves based on AUC values derived from predicted probabilities.

Results

Study Population

We collected 61 volumetric OCT scans (50 AMD eyes and 11 healthy control eyes) obtained from 50 participants (43 with AMD and 7 healthy control), aged 77.65 ± 9.24 years (mean ± SD), of whom 72% were women. The training set included 33 participants, with 27 contributing one eye (27 scans) and 6 contributing both eyes (12 scans), totaling 39 scans, which were divided into 35 for training and 4 for validation. The testing set comprised 17 participants (10 with AMD and 7 healthy controls) contributing 22 scans (11 with AMD and 11 healthy controls).

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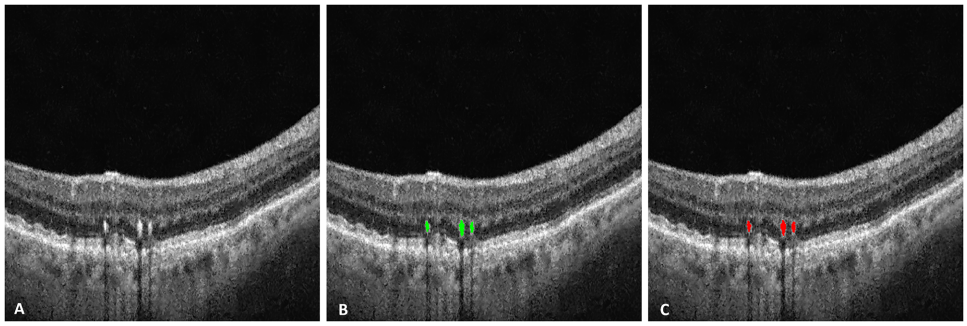


Figure 3. Representative example of automated hyperreflective foci (HRF) segmentation. (**A**) Input OCT B-scan. (**B**) Ground truth HRF annotations (*green*) overlaid on the OCT B-scan. (**C**) Predicted HRF segmentation maps (*red*) generated using Foci-Net.

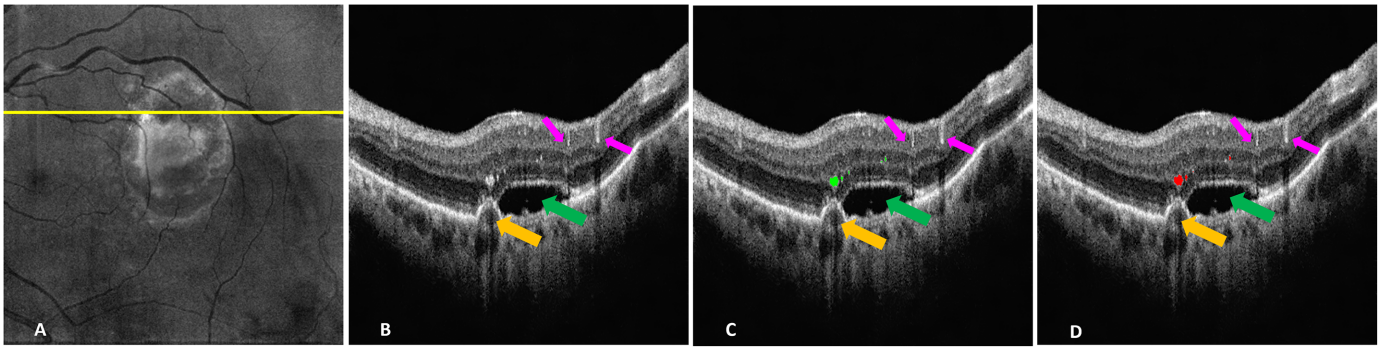


Figure 4. Automated hyperreflective foci (HRF) segmentation in exudative age-related macular degeneration case. (**A**) The en face OCT, with a *yellow line* showing the OCT B-scan position. (**B**) OCT B-scan with drusen (*orange arrow*), subretinal fluid (SRF) (*green arrow*), and blood vessels (*purple*). (**C**) Ground truth HRF annotations (*green*) overlaid on the OCT B-scan, and (**D**) automated HRF segmentation maps (*red*) generated using Foci-Net, with drusen (*orange arrow*), SRF (*green arrow*), and blood vessels (*purple*).

Qualitative Performance

To qualitatively evaluate the segmentation performance of Foci-Net, we applied the trained model to the whole test set. We examined a representative example, as shown in Figure 3, where the model produced accurate HRF segmentations. In Figure 3C (red), the HRF predictions generated by Foci-Net closely match the expert annotations (see Fig. 3B, green) in the OCT B-scan. These qualitative results highlight the model's ability to identify HRF in OCT B-scans reliably.

HRF Segmentation Amidst Pathologies and Vascular Structures

Segmentation of biomarkers within anatomically similar structures is a challenging task.[24] We demonstrated the effectiveness of our model (Foci-Net) in segmenting HRF in complex cases. Figures 4, 5, and 6 illustrate HRF segmentation in exudative and non-exudative cases, and eyes with pigment epithelial detachment (PED), where similar structures such as blood vessels, along with pathologies like drusen, and subretinal fluid (SRF) coexist. This capability is likely due to Foci-Net's ability to capture both local and global context within the image and the inclusion of B-scans without HRF.

HRF 2D and 3D Visualization

To better demonstrate the spatial distribution of HRF, we applied Foci-Net to each B-scan and generated 2D and 3D HRF segmentation, as shown in Figure 7. The 2D HRF projections on en face OCT images, generated from the inner nuclear layer (INL)/outer plexiform layer (OPL) to ellipsoid zone (EZ) (inner segment [IS]/outer segment [OS]) slab, illustrate the spatial distribution and accuracy of Foci-Net predictions compared with manual annotations. These projections include both ground truth and predicted segmentation, demonstrating strong agreement between the model and expert annotations. Clinically, en face images are useful because they offer a clear and intuitive view for comparing the location and extent of HRF across the retina. An example comparison between ground truth and predictions is presented in Figures 7A and 7B. Additionally, we show the 3D volumetric HRF distribution (see Fig. 7C). Compared

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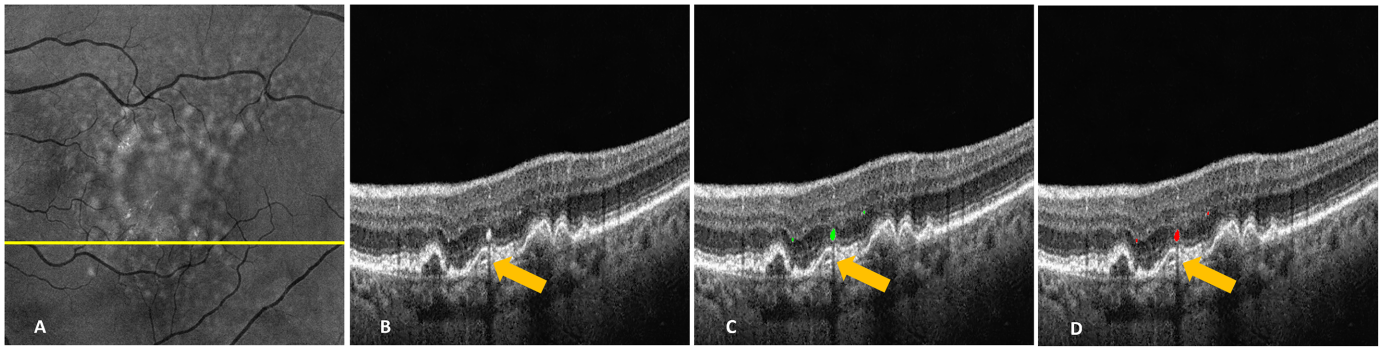


Figure 5. Automated hyperreflective foci (HRF) segmentation in non-exudative age-related macular degeneration case. (**A**) The en face OCT, with a *yellow line* showing the OCT B-scan. (**B**) OCT B-scan with drusen (*orange arrow*). (**C**) Ground truth HRF annotations (*green*) overlaid on the OCT B-scan, and (**D**) Automated HRF segmentation maps (*red*) generated using Foci-Net.

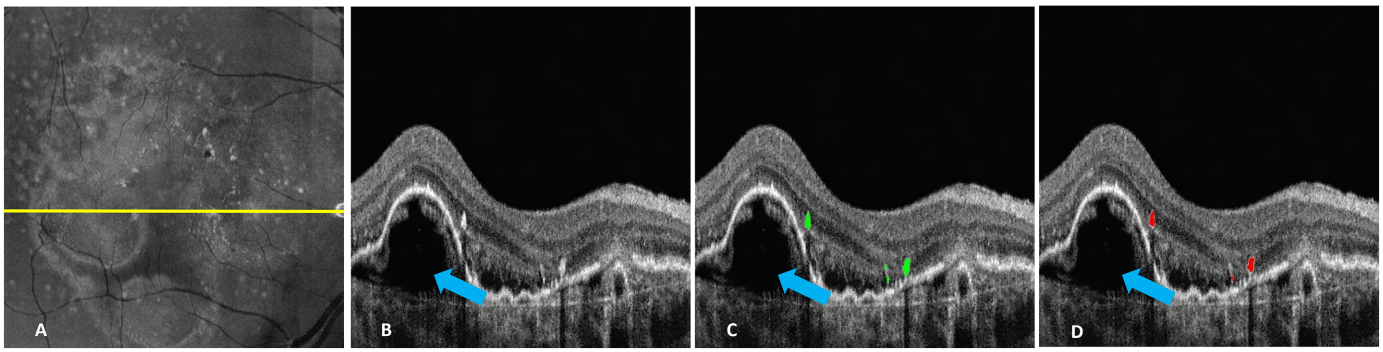


Figure 6. Automated hyperreflective foci (HRF) segmentation in OCT B-scan of a retinal pigment epithelial detachment (PED) case. (**A**) The en face OCT, with a *yellow line* showing the OCT B-scan position. (**B**) OCT B-scan with PED (*light blue arrow*). (**C**) Ground truth HRF annotations (*green*) overlaid on OCT B-scan, and (**D**) automated HRF segmentation maps (*red*) generated using Foci-Net.

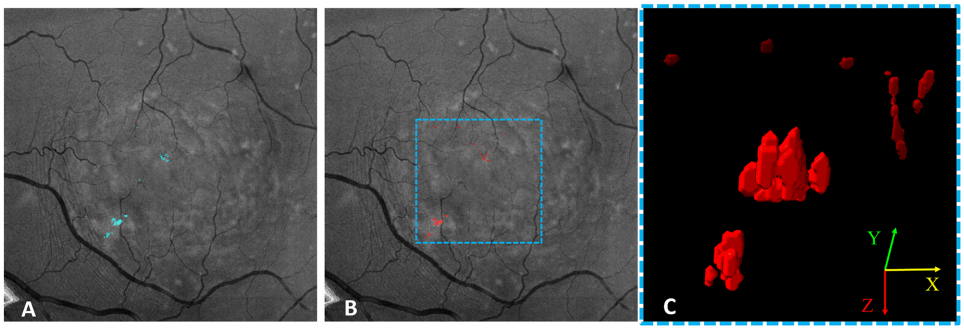


Figure 7. Two-dimensional (2D) and 3-dimensional (3D) visualization of hyperreflective foci (HRF). (**A**) The en face OCT with ground truth HRF annotations (*cyan*). (**B**) The en face OCT HRF segmentation maps (*red*) generated using Foci-Net; the *blue dashed* region indicates the area visualized in 3D in panel **C**. (**C**) The 3D visualization of HRF (*red*).

to 2D, the 3D visualization provides a clearer and more detailed assessment of HRF distribution. The 2D en face HRF area can be misleading, as visually similar regions may correspond to different 3D HRF volumes due to loss of depth information. In contrast, 3D visualization enables more accurate and meaningful quantification of HRF distribution.

Quantitative Evaluation

Segmentation Metrics

Table 1 summarizes the quantitative evaluation results, comprising two levels of performance analysis: (1) count-level, based on 2D en face HRF projection maps derived from the entire OCT volume,

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Table 1. Accuracy of Hyperreflective Foci Segmentation at Count and Pixel Levels (Three-Fold Cross-Validation)

Model	Performance Type	F1-Score (%)	Precision (%)	Sensitivity (%)
Foci-Net	Count-level	73.0 ± 3.8	70.0 ± 7.4	76.7 ± 2.7
	Pixel-level (B-scan)	71.6 ± 29.1	77.7 ± 30.1	71.0 ± 31.5

Table 2. Accuracy of Hyperreflective Foci Classification on Eye and B-scan Levels (Three-Fold Cross-Validation)

Model	Classification Type	Sensitivity (%)	Specificity (%)	AUC (%) (95% CI)
Foci-Net	Eye level	100 ± 0.0	84.9 ± 5.2	100 ± 0.0 (100–100)
	B-scan level	89.4 ± 4.9	89.0 ± 7.0	92.7 ± 0.3 (91.1–94.1)

AUC, area under the curve; CI, confidence interval.

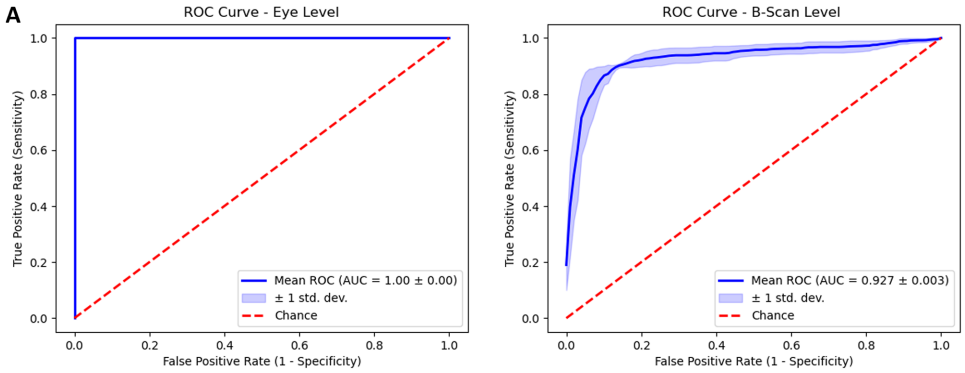


Figure 8. Receiver operating characteristic (ROC) curve from three-fold cross-validation for eye and B-scan level classification. (**A**) Foci-Net (left eye level, right B-scan level).

and (2) pixel-level, based on B-scan segmentations. Foci-Net achieved a mean F1-score of 73.0 ± 3.8 for count-level and 71.6 ± 29.1 for pixel-level segmentation.

Classification Metrics

We performed classification at both the eye level and B-scan level (using 3-fold cross-validation) to evaluate the model's ability to detect HRF across different clinical scenarios. At the eye level, the model was evaluated on its ability to distinguish AMD eyes from healthy controls based on the overall presence or absence of HRF within the volume. For the B-scan level, the model identified individual B-scans within AMD volumes that contained HRF. Thus, using a two-level classification approach captures global (volume-level) and localized (B-scan-level) detection performance. The results, summarized in Table 2 and illustrated in the ROC curve of Figure 8, demonstrate the model's effectiveness in identifying HRF as a relevant imaging biomarker.

Ablation Study

Segmentation (Count and Pixel Level)

The segmentation performance of Foci-Net was evaluated through an ablation study, highlighting its enhanced feature extraction and improved segmentation accuracy. Table 3 summarizes the quantitative results comparing Foci-Net, Foci-Net without the FCFE block, and U-Net, using three-fold cross-validation. Foci-Net count and pixel-level segmentation achieved an average F1-score of 73.0 ± 3.8 and 71.6 ± 29.1, respectively, showing more acceptable performance compared to the Foci-Net without the FCFE block and U-Net.

Foci-Net demonstrated strong performance in segmenting both small and large hyperreflective foci (Fig. 9B), outperforming the baseline U-Net (Fig. 9C) and a variant without the FCFE block (Fig. 9D). The inclusion of the FCFE block enables multiscale feature extraction, which improves the model's ability to detect foci of varying sizes. This design (see Fig. 2C) choice enhances segmentation accuracy by helping the model

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Table 3. Accuracy of Hyperreflective Foci Segmentation at Count and Pixel Levels in the Ablation Study (Three-Fold Cross-Validation)

Models	Performance Type	F1-Score (%)	Precision (%)	Sensitivity (%)
Foci-Net	Count-level	**73.0 ± 3.8**	70.0 ± 7.4	76.7 ± 2.7
	Pixel-level (B-scan)	**71.6 ± 29.1**	**77.7 ± 30.1**	**71.0 ± 31.5**
Foci-Net without the FCFE block	Count-level	72.3 ± 2.1	**75.5 ± 5.9**	69.5 ± 5.4
	Pixel-level (B-scan)	64.7 ± 32.9	73.3 ± 33.9	63.7 ± 35.4
U-Net	Count-level	64.8 ± 7.7	57.1 ± 12.9	**77.4 ± 4.8**
	Pixel-level (B-scan)	67.5 ± 29.9	70.9 ± 30.3	70.9 ± 33.3

FCFE, fine-to-coarse feature extraction.
Bold face figures indicate best performance.

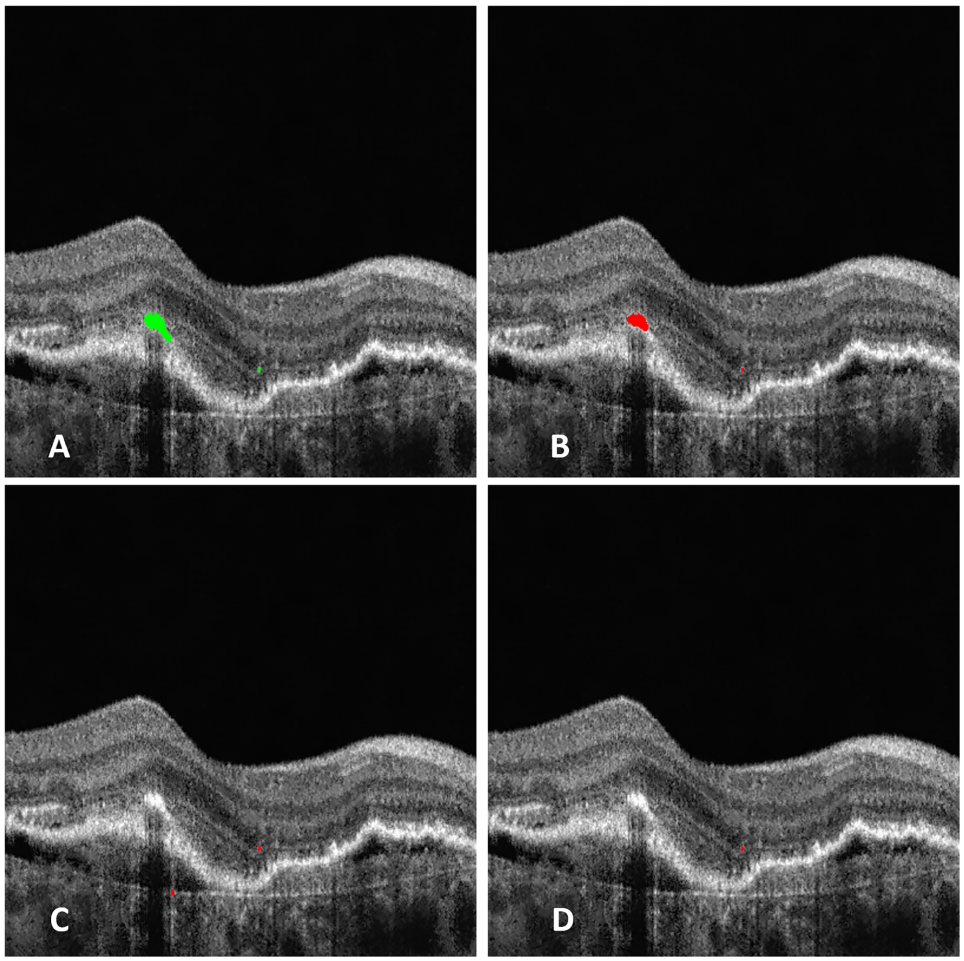


Figure 9. Qualitative comparison of U-Net and its variants. (**A**) Ground truth hyperreflective foci (HRF) annotations (*green*) overlaid on OCT B-scan. (**B**) HRF segmentation maps (*red*) generated using Foci-Net. (**C**) HRF segmentation maps (*red*) generated using a U-Net. (**D**) HRF segmentation maps (*red*) generated using Foci-Net without the fine-to-coarse feature extraction block.

differentiate between similar anatomical structures and reducing false segmentations.

Classification (Eye and B-Scan Levels)

To further evaluate model performance, we conducted an ablation study comparing three architectures: (1) Foci-Net, (2) Foci-Net without the FCFE block, and (3) U-Net. Foci-Net demonstrated superior performance at both the eye and B-scan level of classification tasks, indicating the effectiveness of its architectural enhancements in learning discriminative features for HRF-based diagnosis. As shown in Table 4 in bold face, Foci-Net outperformed the

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Table 4. Accuracy of Hyperreflective Foci (HRF) Classification on Eye and B-scan Levels Using Ablation Study (Three-Fold Cross-Validation)

Model	Classification Type	Sensitivity (%)	Specificity (%)	AUC (%) (95% CI)
Foci-Net	Eye level (11 healthy eyes, 11 AMD eyes)	100 ± 0.0	**84.9 ± 5.2**	**100 ± 0.0** (100–100)
	B-scan level (11 AMD eyes)	89.4 ± 4.9	89.0 ± 7.0	**92.7 ± 0.3 (91.1–94.1)**
Foci-Net without the FCFE block	Eye level (11 healthy eyes, 11 AMD eyes)	100 ± 0.0	78.8 ± 13.9	98.6 ± 1.9 (94.9–100)
	B-scan level (11 AMD eyes)	85.3 ± 5.3	**91.3 ± 5.4**	92.3 ± 0.5 (90.8–93.7)
U-Net	Eye level (11 healthy eyes, 11 AMD eyes)	100 ± 0.0	40.0 ± 27.3	91.7 ± 6.4 (76.9–100)
	B-scan level (11 AMD eyes)	**90.9 ± 3.9**	84.5 ± 9.9	90.2 ± 0.8 (88.4–91.8)

AUC, area under the curve; AMD, age-related macular degeneration; CI, confidence interval; FCFE, fine-to-coarse feature extraction.

Bold face figures indicate best performance.

comparison models across both eye and B-scan level classification tasks. This performance is reflected in the ROC curves in Figure 8A, compared to the alternative model's ROC curves in Figures 10A and 10B, with Foci-Net achieving an AUC of 100 ± 0.0% for the eye and 92.7 ± 0.3% for the B-scan level. This classification enables the automated identification of eyes and B-scans containing HRF, which may be clinically relevant to the severity and progression of AMD. Foci-Net demonstrated significantly higher AUC compared with both the variant without the FCFE block ($P = 0.013$) and the U-Net baseline ($P < 0.001$).

Volume Distribution of HRF in Retinal Layers

We quantified the proportional volume distribution of HRF in 50 volumetric OCT scans in cases with AMD. The total HRF volume across all retinal layers, as illustrated in Figure 11A, was obtained, and then we calculated the HRF proportion in each retinal layer. Out of the total HRF volume across all retinal layers, the highest HRF volume was in the outer nuclear layer (ONL), followed by OPL, EZ-RPE and INL (see Fig. 11).

Inference Time

On average, the proposed model required approximately 62.5 milliseconds per B-scan and 32 seconds per volumetric OCT scan. This short inference time enables near-real-time B-scan and volumetric processing and could be suitable for integration into clinical OCT settings.

Discussion

In this study, we presented Foci-Net, a deep CNN designed to segment HRF in AMD. Foci-Net performance was evaluated using 3-fold cross-validation, achieving an average F1-score of 73.0% for count-level (en face) and 71.6% for pixel-level (B-scan) segmentation relative to the ground truth. Foci-Net demonstrated robust performance, achieving 100 ± 0.0 AUC at the eye level and 92.7 ± 0.3 at the B-scan level. Furthermore, the quantification of HRF volume revealed that the highest volume was located in the ONL (55.37%), as shown in Figures 11B and 11C.

This study presents several key contributions as follows:

First, the introduction of the FCFE block improved Foci-Net's segmentation accuracy for HRF of varying sizes (see Fig. 9). The effectiveness of multi-scale feature extraction using dilated convolutions has been demonstrated in prior work. For example, Zhang et al.[25] showed that incorporating dilated convolutions and multi-scale residual modules into the U-Net enhanced feature representation and reduced information loss in brain tumor segmentation. Similarly, Liu et al.[26] reported that a pyramid dilated convolution module improved segmentation accuracy for skin lesions of varying sizes by capturing multi-scale contextual information. In our model (Foci-Net), we utilize the dilated convolution concept in the network's bottleneck to capture multi-scale information, such as fine, mid, and global contextual features. To confirm the effectiveness of our FCFE module, we assessed the risk of overfitting and conducted ablation studies as part of our investigation. Ablation studies confirmed the effec-

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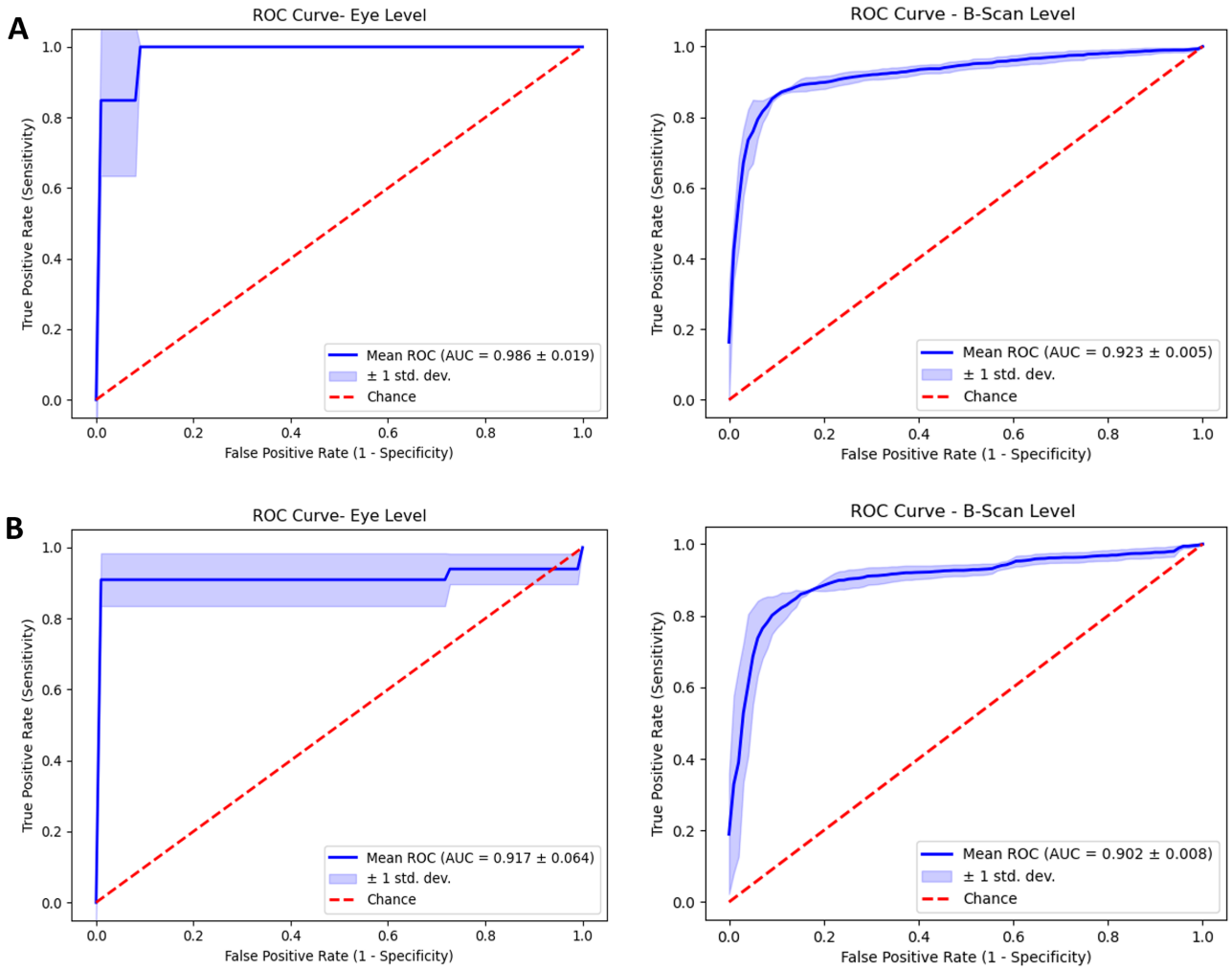


Figure 10. Receiver operating characteristic (ROC) curve from three-fold cross-validation for eye and B-scan level classification using ablation study. (**A**) Foci-Net without the fine-to-coarse feature extraction block. (**B**) U-Net.

tiveness of the FCFE block, demonstrating that Foci-Net outperformed both the variant without FCFE and the baseline U-Net in HRF segmentation and classification tasks.

Second, previous studies[9,12,13] segmented HRF at the B-scan level but did not explicitly quantify or analyze HRF at the en face or 3D volumetric level. In contrast, we utilize B-scan segmentations to enable layer-specific HRF quantification with en face and 3D analysis. The reason behind training on B-scans allows our model to be directly applicable to conventional commercially available structural OCT systems that acquire B-scans and less dense volumetric OCT scans, which are widely used in routine clinical practice. This maximizes the potential for real-world clinical deployment. Another reason is that B-scans-based training enables the use of a larger number of samples per volume to improve data efficiency. Each B-scan serves as a separate training sample, allowing faster convergence during training. Compared with 3D CNNs, 2D CNN models trained on B-scans are significantly lighter in terms of memory and computational requirements. Although 3D CNNs can leverage volumetric spatial information, training on a small set of OCT volumes greatly increases the risk of overfitting, limiting their generalizability and clinical deployment. Additionally, we quantified HRF volume in multiple retinal layers, as illustrated (see Fig. 11). We found that HRF volume is mainly located in the ONL in eyes with AMD, an expected finding given that the disease primarily affects photoreceptors and the RPE. This ability to automatically and accurately quantify HRF may provide additional information for clinicians to understand the extent of disease and be useful for monitoring disease progression.

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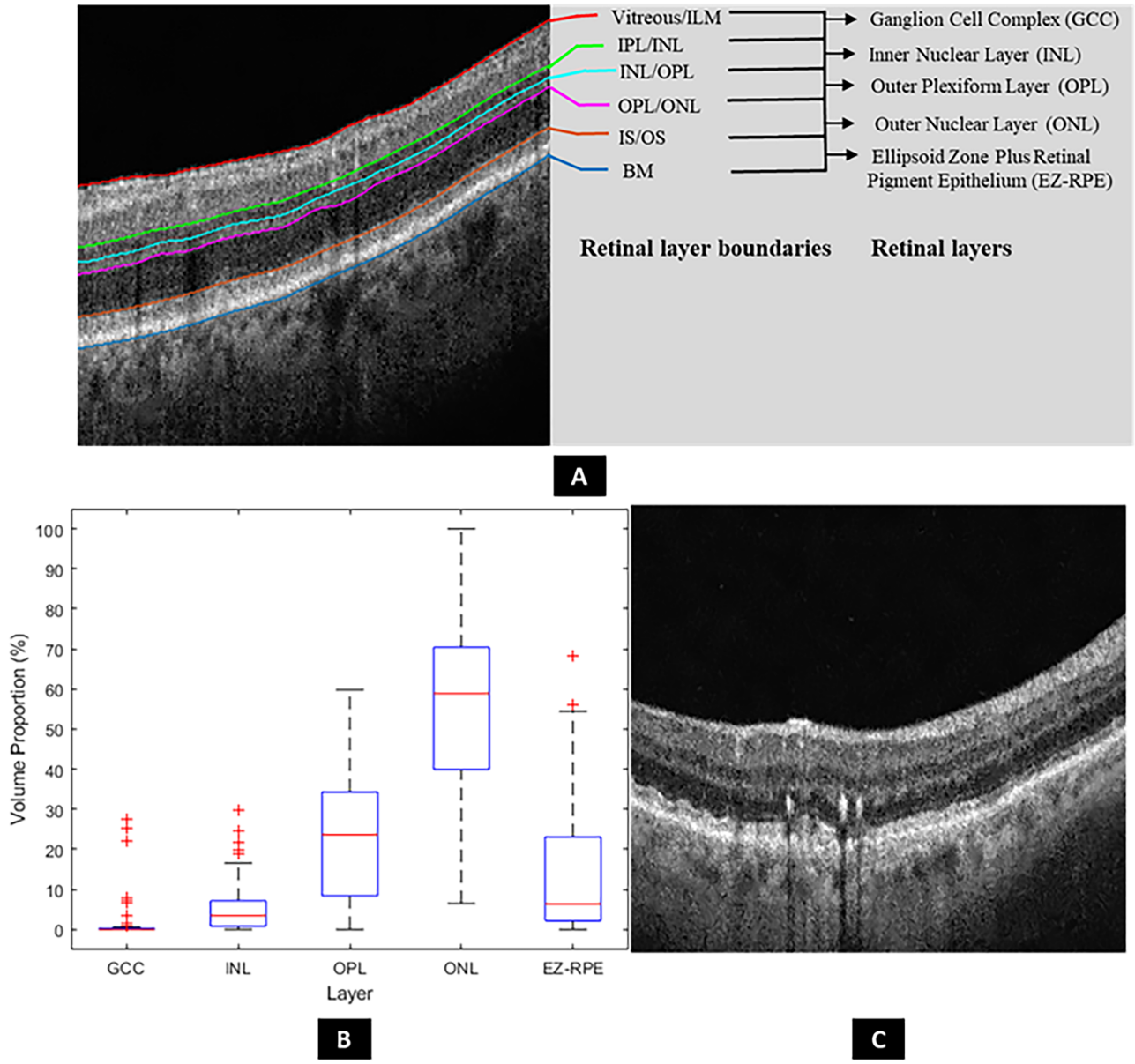


Figure 11. Volume distribution of hyperreflective foci (HRF) in retinal layers. (**A**) OCT B-scan showing retinal layer boundaries (inner limiting membrane [ILM], inner plexiform layer [IPL], inner nuclear layer [INL], outer plexiform layer [OPL], outer nuclear layer [ONL], inner segment [IS], outer segment [OS], and Bruch's membrane [BM]). (**B**) Proportional HRF volume distribution in retinal layers. (**C**) An example of a B-scan shows most HRFs are in the ONL.

Third, we presented 3D volumetric representations of HRF, as illustrated in Figure 7. Whereas 2D en face maps show the spatial distribution of HRF across the retinal surface, they lack depth context. In contrast, the 3D preserves axial information, and the visualization offers a more comprehensive and intuitive view of HRF. This form of representation is important for clinicians, allowing them to assess the HRF location and how deeply it is inside the retina, as well as how it evolves through different layers. The 3D representation is important in AMD, where studies have revealed that HRFs are clinical biomarkers for AMD progression and could predict the progression from iAMD to the development of late-stage AMD.[4,6,27] As imaging speed and resolution in OCT systems continue to improve,[28] with more B-scans acquired in less time and minimal motion artifacts, volumetric OCT is expected to become standard in both research and clinical settings. Although B-scan-based analysis remains useful and computationally efficient, volumetric HRF quantification could support risk stratification, longitudinal monitoring, and response to treat-

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ment.[29] Our present work serves as a foundation for future developments that aim to bring deep learning-based enhanced OCT analysis into clinical practice.

Finally, we demonstrated the capability of Foci-Net in both B-scan and eye-level classification, effectively distinguishing B-scans with and without HRF. This feature is particularly important for commercial OCT systems, where patients are typically imaged using only a few cross-sectional OCT B-scans. Similarly, eye-level classification is clinically valuable for differentiating between eyes with and without HRF. These capabilities highlight the potential utility of our model in the early screening of B-scans and eyes with HRF. Prior studies have shown that the presence of HRF is associated with reduced cone-mediated function (responsible for central, high-resolution, and color vision), suggesting that HRF is an early marker of functional vision loss and AMD severity.[30] Importantly, early identification of high-risk features like HRF has been linked to progression from early to late-stage AMD.[31]

Two trained graders generated HRF ground-truth labels by independently reviewing B-scans, with joint discussion limited to questionable cases to reach a final, agreed-upon label. Because we only saved the final agreed labels, inter-grader agreement metrics, such as Dice or ICC, were not calculated. We acknowledge this as a limitation and note it as an important direction for future work.

Currently, Foci-Net segments HRF using binary classification approach. We plan to upgrade to a multi-class approach to distinguish HRF types, potentially offering deeper insights into disease progression and improving clinical relevance. This multiclass extension would involve annotating HRF based on morphological characteristics, such as size, reflectivity, and the presence or absence of back-shadowing, followed by training a multiclass segmentation network. HRF is observed in a variety of retinal diseases (e.g. AMD, DR, and DME), and their definition and clinical significance vary across conditions. Literature reports demonstrated that HRF may represent different pathological abnormalities, including activated microglial cells, hard exudates, migrating RPE cells, neuroinflammatory cells, and lipid deposits leaking from damaged vasculature.[32] Clinically, it is important to determine whether these foci are localized to specific retinal layers in different AMD subtypes. For example, comparing HRF distribution between geographic atrophy and macular neovascularization could provide valuable insights into disease mechanisms and progression. Given these diverse pathophysiological origins, tracking HRF based on their morphological characteristics is crucial for assisting clinicians in the diagnosis and management of retinal disease. Therefore, training a model to differentiate HRF subtypes could enhance understanding of their clinical relevance, severity, progression, and response to treatment. Model performance may change in larger, multi-center cohorts due to differences in imaging devices and disease subtypes. Future validation on larger, multi-center datasets will be essential to evaluate generalizability and robustness across clinical settings.

In conclusion, the proposed model enables accurate automated segmentation of HRF. The algorithm demonstrated its ability to segment HRF in complex pathological cases, where accurate discrimination of HRF from similar-appearing structures is essential.

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